Data Engineering

Spring 26 – DS471

Jan 20, 2026

Logistics

Class timings	Mon, Tues 11:30 PM – 12:20 PM, LH6, Fri 11 am
Credit Hours	2 hours + 1 hour (Practical)
Office Hours	Monday, Tuesday 10 am – 11 am, S18 2nd Floor NAB or by appointment
MS Teams:	TBD
Contact:	MS Teams, Email (farah.saeed@giki.edu.pk), WhatsApp

Instructor Details

Farah Saeed

- **BS:** FAST Isb, 2016
- **PhD:** UGA US, 2025 (Expected)
- **Research:** Applying AI in science and environment

Data Engineering – Course Introduction

This course introduces the world of **Data Engineering**—the backbone of modern data-driven systems. You'll learn how raw data is collected, cleaned, moved, and transformed into reliable data pipelines that power analytics, dashboards, and AI systems.

Through hands-on work with **SQL, Python, Spark, Airflow, Kafka, and open source platforms**, you can design and build scalable data pipelines for real-world datasets.

The course focuses on practical skills, smart design choices, and modern tools for working with **big data** at scale.

Attendence

- Per Institute policy
- 80% attendence
- 6 classes in 2 Cr. Hour course

CLO

CLO #	Course Learning Outcome (CLO)	Bloom's Taxonomy Level	Graduate Attribute (GA)
CLO-1	Explain core data engineering concepts including data models, ETL/ELT pipelines, batch and stream processing, and distributed system architectures.	Understand (L2)	GA-1: Engineering Knowledge / Computing Fundamentals
CLO-2	Analyze real-world data sources and workloads to select appropriate data storage systems and processing paradigms.	Analyze (L4)	GA-2: Problem Analysis
CLO-3	Design scalable, reliable, and cost-effective data pipelines integrating ingestion, transformation, validation, and orchestration.	Create (L6)	GA-3: Design / Development of Solutions
CLO-4	Implement data engineering solutions using modern tools such as SQL, Python, Spark, Airflow, Kafka, and cloud data platforms.	Apply (L3)	GA-5: Modern Tool Usage

Grading (Tentative)

CLOs Assessment Mechanism				
Assessment Tools	CLO_1	CLO_2	CLO_3	CLO_4
Quizzes	0%	20%	0%	0%
Assignments	0%	0%	20%	0%
Project	0%	0%	0%	100%
Mid Term	50%	40%	40%	0%
Final Term	50%	40%	40%	0%

Grading

Assessment Item	Weightage
Quizzes	10%
Assignments	5%
Projects	15%
Mid Term	30%
Final Term	40%

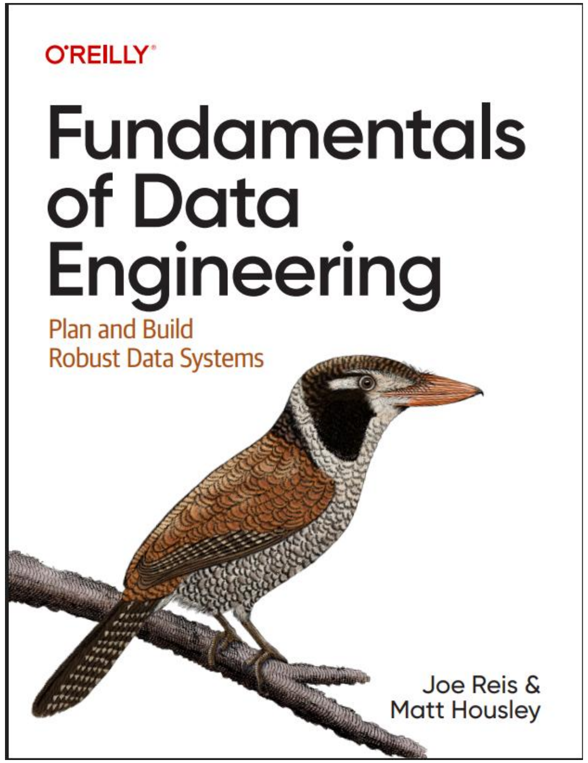
Quizzes

- Total 5 + 2 quizzes.
- Counted also 5.
- 2 retakes.

Assignments

- Mostly individual assignments
- Deduction for late submissions within 7 days.
- Assignment not accepted after 7 days even if there is a genuine reason.
- *If a student informs ahead of time that submission portal is not working then their assignment can be accepted through email. Last modified date is to be checked and has to be before the submission date and time.*

Textbook



Fundamentals of Data Engineering by
Joe Reis and Matt Housley

+ Mix of verified sources from web.

Referenced

Why Data Engineering Matters

Companies generate **massive amounts of data every second**

- Bad data pipelines = bad decisions
- AI, dashboards, and ML models **fail without good data**

- **No pipelines → No analytics → No AI**

- Data engineers are the **unsung heroes** behind:
  - Netflix recommendations
  - Google search
  - Real-time fraud detection
  - Health and climate analytics and so on.

Smaller dataset

Number of influenza like illness cases on weekly basis from 2010 to 2024 in US

date	ILITOTAL	OT
10/4/2010	8473	746485
10/11/2010	9347	746230
10/18/2010	9684	777397
10/25/2010	9904	781234
11/1/2010	11020	766753
11/8/2010	11807	769982
11/15/2010	13336	787038
11/22/2010	12079	606527
11/29/2010	13997	772016
12/6/2010	14958	742606
12/13/2010	18022	704923
12/20/2010	19899	588406
12/27/2010	20346	592880
1/8/2011	20026	744043
1/10/2011	20721	711908
1/17/2011	26257	751850
1/24/2011	31975	798637
1/31/2011	34284	772974
2/7/2011	34456	812230
2/14/2011	36567	831600
2/21/2011	31277	801449
2/28/2011	24627	781629
3/7/2011	20377	747993
3/14/2011	17309	741807
3/21/2011	14031	728883
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< 700 rows

CDC

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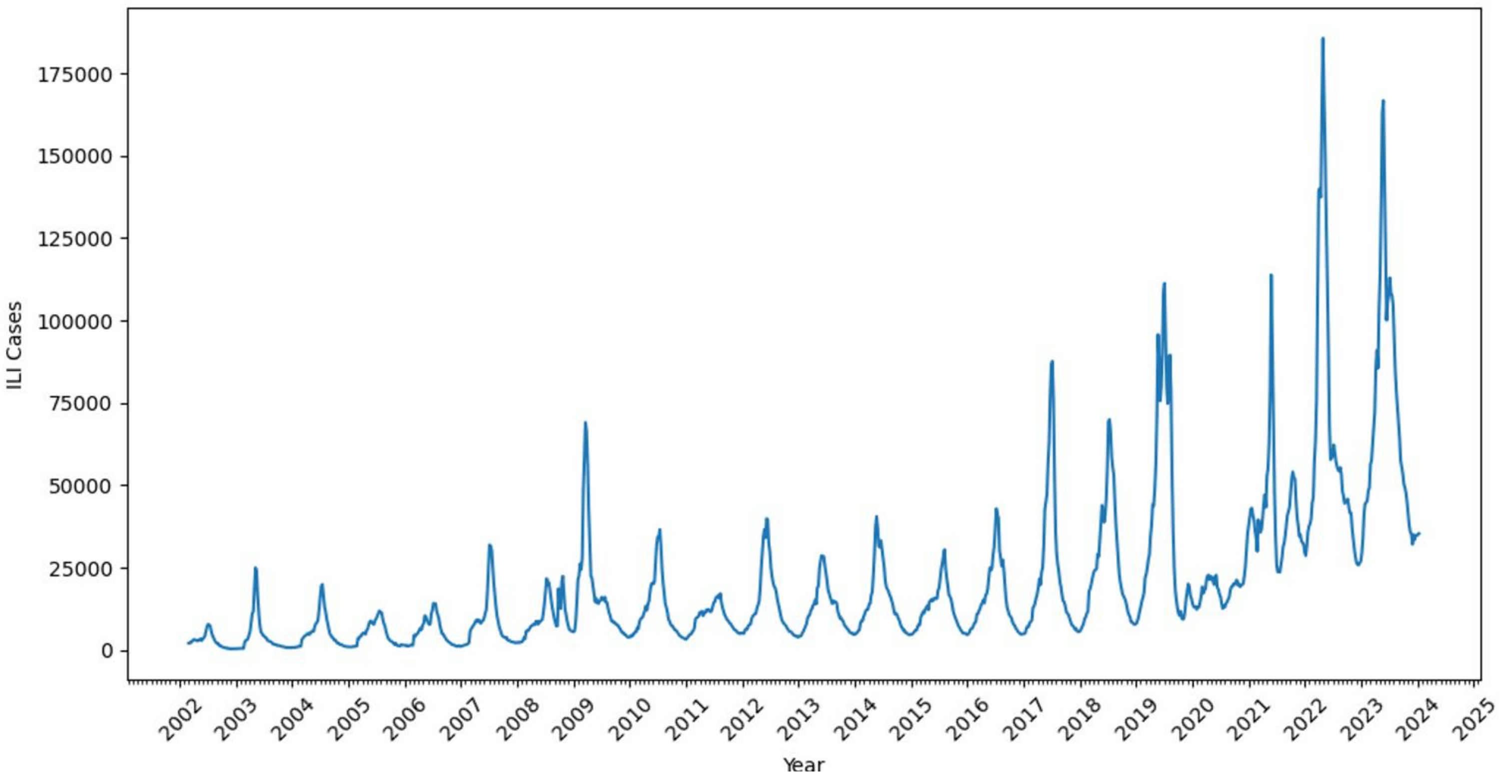

CDC

Why Data Engineering Matters

Data Explosion

Worldwide **data created and stored** is projected to reach *181 zettabytes* by 2025 and *221 zettabytes* by 2026 — enormous scales far beyond petabytes

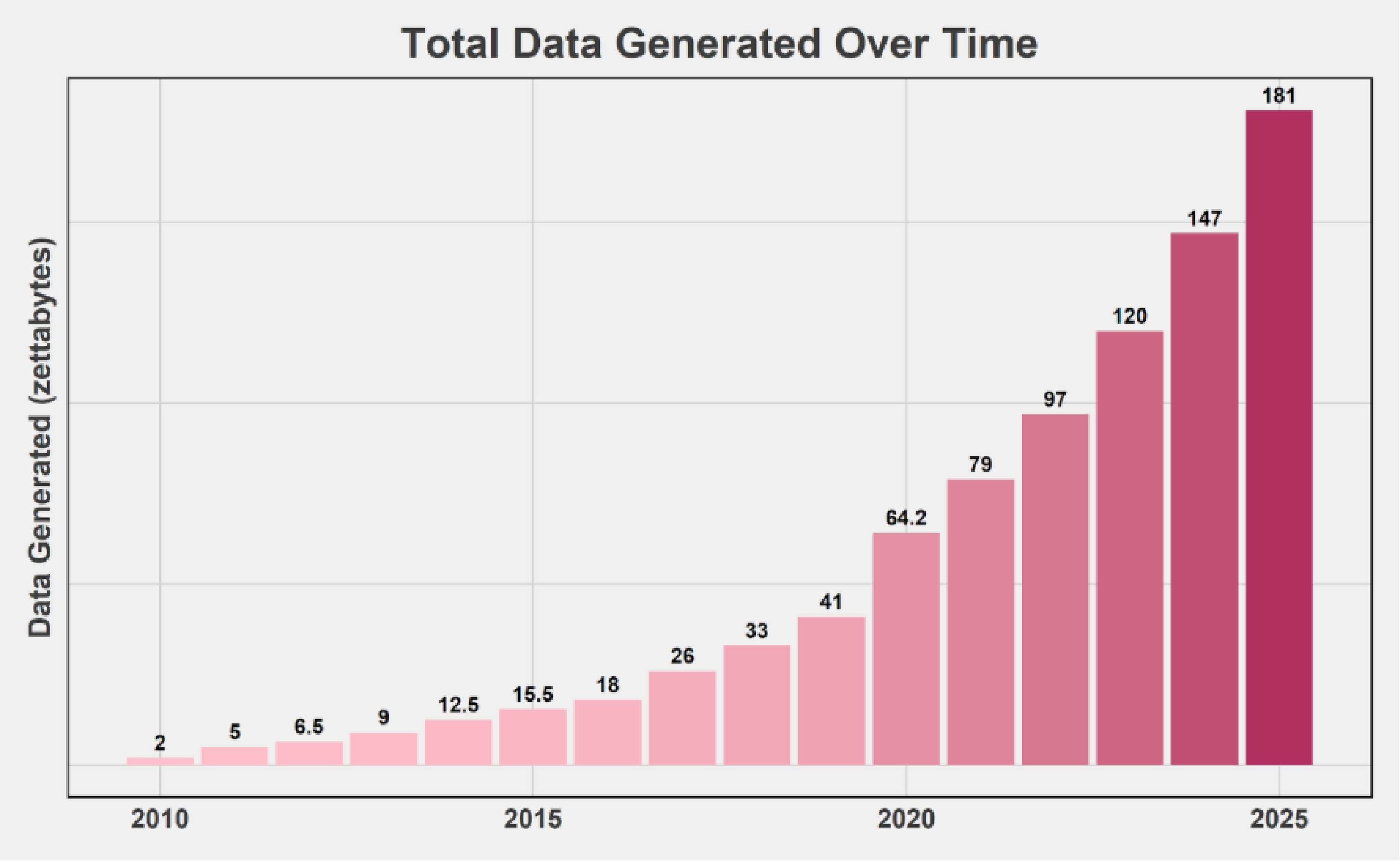

Figure 2: Total data generated over time [source]

https://futuretech.mit.edu/news/what-drives-progress-in-ai-trends-in-data
https://www.statista.com/statistics/871513/worldwide-data-created/

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Every day on the internet:

- **4 PB of data** created on Facebook (petabytes from user activity).
- **500 hours of video per minute** uploaded to YouTube (~1.5 TB/min) — *terabytes every minute*.
- **Hundreds of terabytes of WhatsApp, Instagram, and Snapchat data** generated daily.

This illustrates that today's data is **not just big — it's unimaginable** in human terms.

https://www.sales-hacking.com/en/post/statistiques-big-data?utm_source=chatgpt.com

Realtime Twitter data

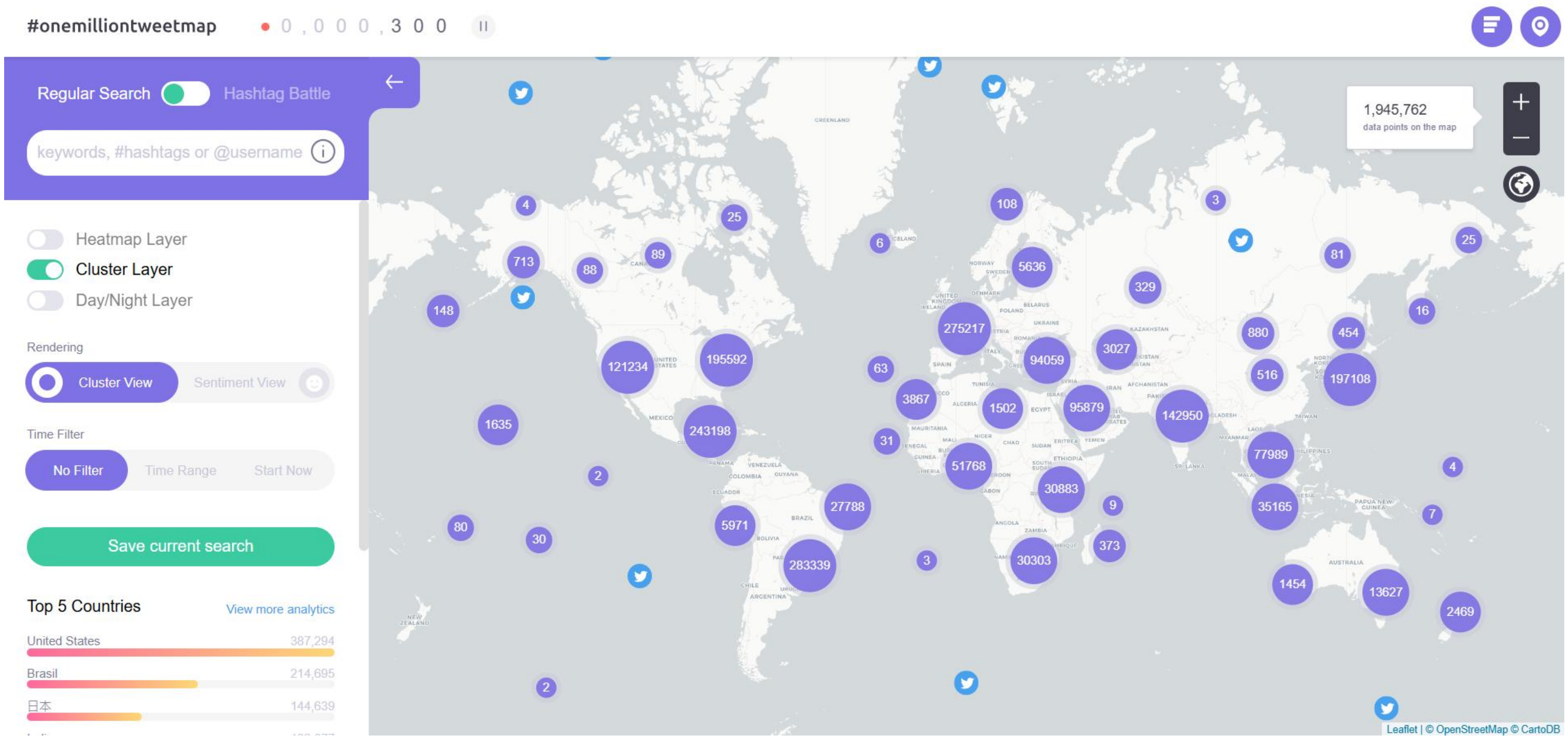
Raw Twitter data consists of billions of tweets — but this visual instantly shows where activity is happening

Tweetping, drops a bright pixel at the location of every tweet in the world

https://www.wired.com/2013/02/tweetping/

Visualizing One Million Tweet Map

Visualizing the last one million geolocated tweets clustered on a world map.



https://onemilliontweetmap.com/?center=22.224838097714475,-15.820312500000002&zoom=2&search=&timeStep=0&timeSelector=0&hashtag1=&hashtag2=sad&sidebar=yes&hashtagBattle=0&timeRange=0&timeRange=25&heatmap=0&sun=0&cluster=1&rendering=sentiment

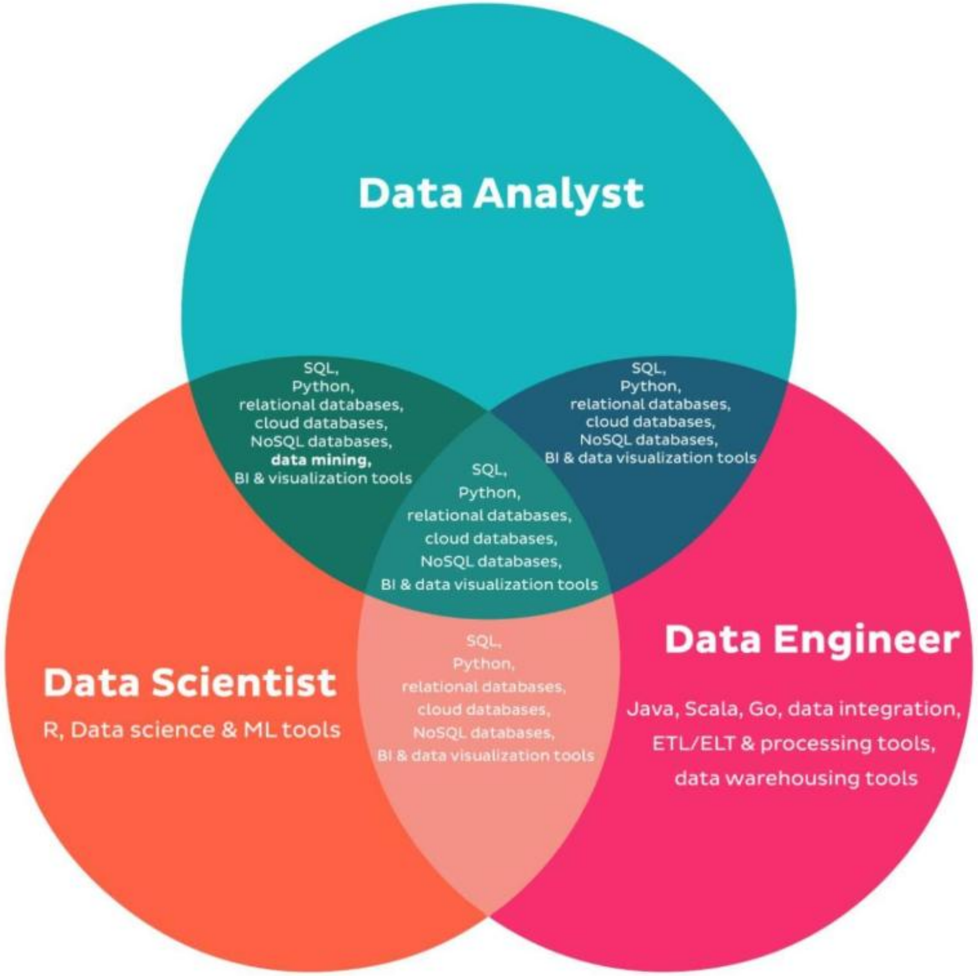
Tweetping, drops a bright pixel at the location of every tweet in the world

What is Data Engineering

Designing and building scalable systems that collect, process, and deliver high-quality data in real time or batches.

Data Engineering makes data usable at scale.

Importance of Data Engineering



https://www.kdnuggets.com/wp-content/uploads/rosidi_navigating_data_science_job_titles_data_analyst_data_scientist_data_engineer_5.jpg

Tools

- SQL – querying data
- Python – data processing
- Spark – big data processing
- Airflow – workflow orchestration
- Kafka – real-time streaming